Grouping Enzymatic Reactions by EC Number

Benchmarking reaction embeddings & clustering — and a fusion method that beats them

13,721 reactions · **22** embeddings × 9 clustering algorithms · EC-purity & retrieval

Goal. Given ModelSEED reactions, find the (reaction embedding, clustering algorithm) pair that groups reactions so each cluster is *clean* with respect to its **EC number** — same enzyme class together, different classes apart.

Why it matters

- EC **annotation transfer** to unlabelled reactions
- **Curation**: flag EC mis-assignments (outliers in a clean cluster)
- A reaction **similarity index** (FAISS) for metabolic modelling

Dataset	
reactions	13,721
full EC (4-level)	5,622
EC3 subclasses	261
EC2 / EC1	68 / 7

non-obsolete, full EC, all compounds have SMILES.

Primary metric: **EC3 AMI** (chance- & count-corrected). Also EC1/EC2/full, homogeneity, purity, and kNN retrieval.

Embeddings (22), by representational family

Structural (diff): ECFP2/4/6, FCFP4, AtomPair, RDKit, MACCS, MHFP, DRFP, Avalon, torsion

PhysChem: RDKit-217 + Mordred-1613 desc.

Atom-mapped: CGR (linear)

Learned: RXNFP, ChemBERTa, **REXzyme**

Reduced-graph: ErG (pharmacophore)

Graph-GNN: GIN

3D geometry: Uni-Mol

The last four (**REXzyme**, **ErG**, **GIN**, **Uni-Mol**) are genuinely *new axes* — Part III.

Clustering (9)

K-means, Agglomerative (avg / complete / Ward), Birch, **Leiden**, **Louvain**, HDBSCAN, Butina

Evaluation

Common 256-d SVD / cosine space.

Clustering: AMI, homogeneity, V-measure, purity at EC1/2/3/full.

Retrieval: leave-one-out kNN EC accuracy.

One harness, every embedding × every algorithm, scored identically — 800+ configurations.

What it asks

If I tell you a reaction's *cluster*, how much does that reveal its true *EC3 subclass*? **Mutual information** MI measures that overlap; **AMI subtracts the** MI **expected by chance**, so a method is not rewarded for merely splitting into many small clusters.

Definition

For clusters U and EC3 classes V over N reactions:

$$\mathrm{AMI}(U,V) = \frac{MI(U,V) - \mathbb{E}[MI]}{\frac{1}{2}\big(H(U) + H(V)\big) - \mathbb{E}[MI]}$$

$H(\cdot)$ = entropy; $\mathbb{E}[MI]$ = mean MI over random labelings with the same cluster/class sizes (`sklearn` default).

How to read a cell

- **1.0** — clusters match EC3 exactly.
- **0.0** — no better than random.
- **Chance- & count-corrected**: fair to compare 45 clusters (Louvain) with 800 (K-means) in one grid.

What it is *not*

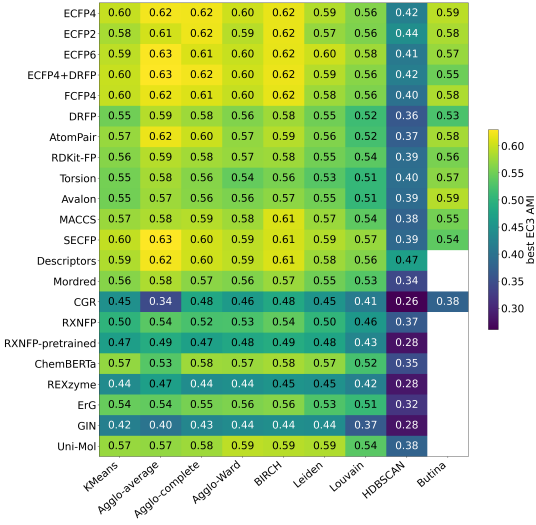
AMI scores an *unsupervised partition*, not prediction accuracy, and on a different scale (tops ≈0.63 here). The number to quote for "can we predict EC?" is the kNN retrieval accuracy (EC3 ≈0.78) — next section.

EC3 AMI = chance-corrected agreement of clusters with EC3 subclasses; **1 = perfect, 0 = random**, comparable across cluster counts.

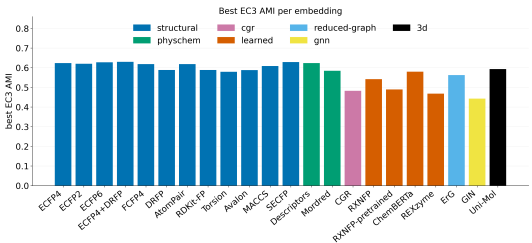
Part I

The benchmark

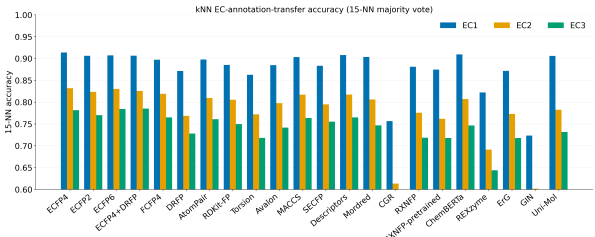
22 embeddings $\times$ 9 algorithms, one scoring harness



CGR rows are full-set (coverage-limited to 75%); the fair CGR comparison is on a later slide.



- **Structural difference** fingerprints form a **plateau** at EC3 AMI ≈0.62–0.63 — radius/type barely matter.
- **PhysChem** (217 descriptors) match them: highest homogeneity/purity.
- **Agglomerative average/complete** + Birch win; graph methods (Leiden/Louvain) coarser; density (HDBSCAN) weakest.
- Learned embeddings underperform (RXNFP worst) — and, spoiler, so do the four *new* axes on the right (Part III).

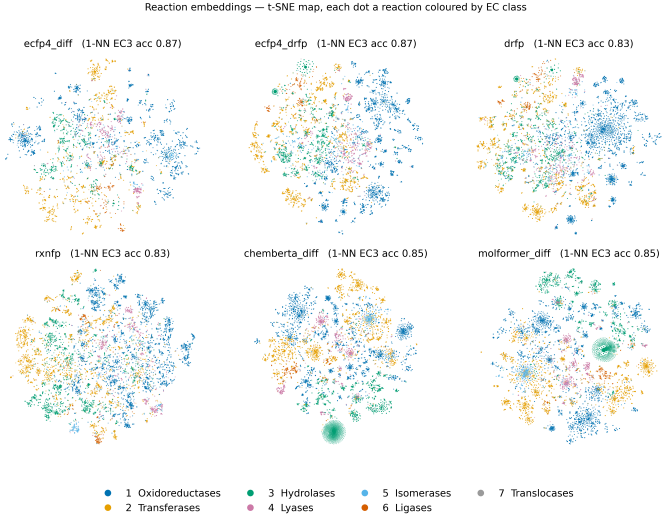


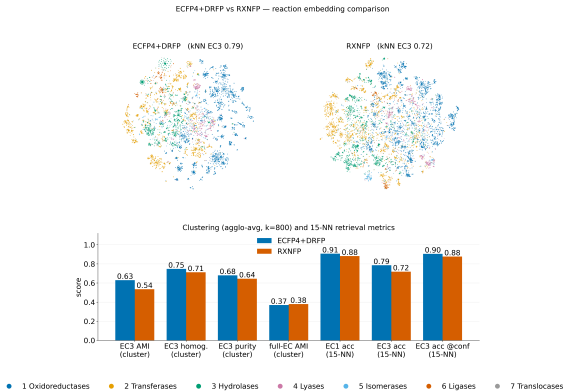
Cluster AMI tops out near 0.63, but **nearest-neighbour retrieval** is far stronger:

- 15-NN: EC1 **0.91**, EC2 0.83, EC3 0.78
- confidence-gated: **EC3 0.90 accuracy over 76%** of reactions

Reading: the representation already knows EC *locally*; unsupervised partitioning loses it *globally*. This is the biggest untapped lever.

For annotation, don't cluster — **retrieve**. Full 4-level EC stays hard everywhere ($\sim$0.3–0.5).


Reaction embeddings — t-SNE map, each dot a reaction coloured by EC class
ecfp4_diff (1-NN EC3 acc 0.87)
ecfp4_drfp (1-NN EC3 acc 0.87)
drfp (1-NN EC3 acc 0.83)
rxnfp (1-NN EC3 acc 0.83)
chemberta_diff (1-NN EC3 acc 0.85)
molformer_diff (1-NN EC3 acc 0.85)
1 Oxidoreductases
2 Transferases
3 Hydrolases
4 Lyases
5 Isomerases
6 Ligases
7 Translocases



RXNFP (pretrained rxn-BERT) trails a cheap fingerprint on every EC metric.

- **Distribution:** trained on USPTO *organic synthesis*, not enzymatic biochemistry.
- **Taxonomy:** fine-tuned for organic reaction-type classes, not EC.
- **No difference operation:** a holistic latent; cofactors dominate.

Adding a difference (`rxnfp_moldiff`) helps only a hair — the ceiling is *training*, not the difference construction. *(Part III revisits this with a biochemistry-native model.)*

Part II

Bond-changes & fusion

the method that beats every single view

Idea. Merge reactant + product graphs into one via **atom mapping**; bonds carry a *dynamic* label (formed / broken / order-changed). It encodes the **actual bond transformation** — exactly what EC classifies.

Pipeline. RXNMapper (atom map) → chython (compose CGR) → fold to fingerprint bits.

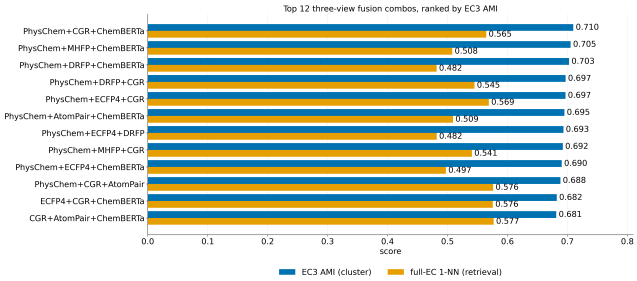
Prior art (Varnek/Nugmanov; Heid & Green 2022); the *new* part is applying it to **EC clustering on ModelSEED** head-to-head.

Fair subset (n=10,260)

	EC3 AMI	**full-EC 1-NN**
ECFP4-diff	0.628	0.490
PhysChem	**0.667**	0.484
CGR	0.613	**0.543**

CGR wins full-EC retrieval (its target!) but doesn't dominate EC3; capped at 75% coverage (hard-to-map enzymatic reactions).

No single representation is best at everything — each view has **complementary** strengths.



Recipe (unsupervised, no fitting): per-view normalize → concat → SVD-256 → agglomerative-average.

All 35 triples from {**PhysChem**, **ECFP4/DRFP/MHFP/AtomPair**, **CGR**, **ChemBERTa**} scored (top 12 shown; bars = EC3 AMI / retrieval accuracy).

- Best EC3 AMI **PhysChem+CGR+ChemBERTa = 0.71** — beats every single view (0.667).
- **CGR**-containing triples win **full-EC retrieval** (0.57–0.58).
- **PhysChem is the anchor** (in every top triple); even "weak" **ChemBERTa** earns its place.

Tight band (0.68–0.71): the *recipe*, not the exact triple, is what matters.

Part III

Do novel representations help?

pharmacophore, learned-graph, 3D shape, biochemistry-native — and why none moves the needle

We asked: does a representation that "sees" the reaction *differently* carry EC signal the 2D fingerprints miss? Four new axes, each a different view:

ErG pharmacophore reduced-graph

RDKit ErG — atoms abstracted to pharmacophore *types* on a reduced topological graph. "Same arrangement of features?"

GIN pretrained 2D-graph GNN

snap-stanford context-prediction GIN — learns on the molecular *graph*, not a SMILES string.

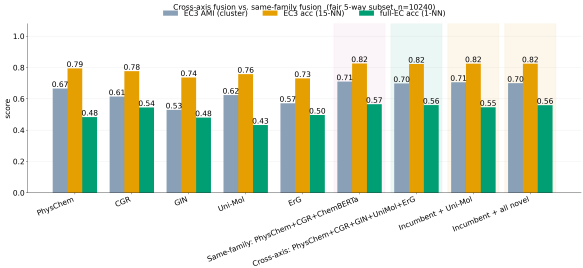
Uni-Mol 3D molecular geometry

SE(3) transformer on generated *conformers* — the only view of true molecular *shape* in the pool.

REXzyme biochemistry-native reaction

335M T5 encoder trained on **32M RHEA reaction→enzyme pairs** — the exact biochemical domain, not USPTO synthesis.

These are real, distinct *representations* — shape, pharmacophore, learned graph, enzyme-supervised language.



Single-view EC3 AMI (fair subset), all under the plateau: **Uni-Mol** 0.62 > **CGR** 0.61 > **ErG** 0.57 > **GIN** 0.53.

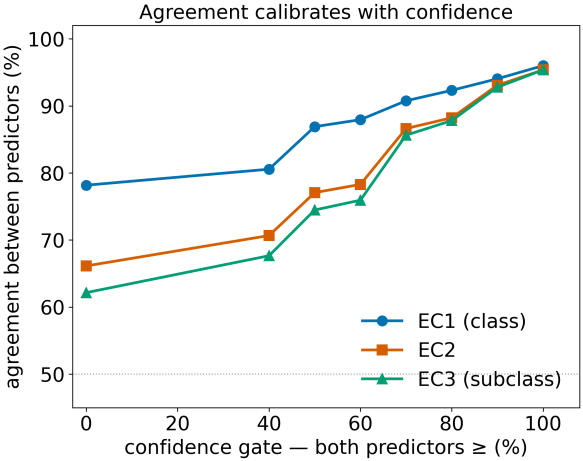
- **REXzyme** headline negative: biochem-native pretraining annotates worse than USPTO rxnfp (0.78 vs 0.83).
- Fusing all four loses to the Part II incumbent (0.698 vs **0.710**); adding the best one on top hurts (-0.004).
- The two most label-aligned *off-the-shelf* FPs also miss: **FCFP4** 0.62 ties ECFP; **Mordred** (1613 desc.) 0.59 < RDKit-217 — more features hurt.

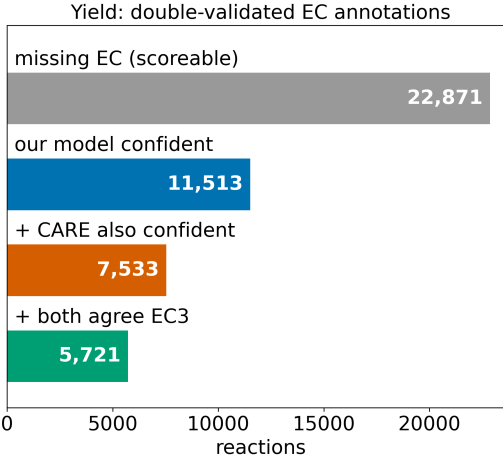
Representational diversity $\neq$ task-relevant diversity.

EC is a **bond-change** label; once descriptors+CGR+ChemBERTa capture that, shape / pharmacophore / graph / catalysis views are redundant. The plateau is the *taxonomy*, not any encoder's ceiling.

Rung	Method	Status / expected
0	Best incumbent (**ECFP**/**CGR**/**PhysChem**) + agglo	EC3 AMI 0.63–0.67
1	**Multi-view fusion** (unsupervised)	**0.71 — done, beats all singles**
–	*novel orthogonal axes* (**3D**/**ErG**/**GIN**/**REXzyme**)	no gain — ruled out (Part III)
2	+ cofactor normalization	de-confound ATP/NAD currency
3	+ self-supervised contrastive	label-free ceiling
4	+ **hierarchical-EC metric learning**	the label lift (biggest lever)
5	rung 4, **held-out EC subclasses**	does it generalize?

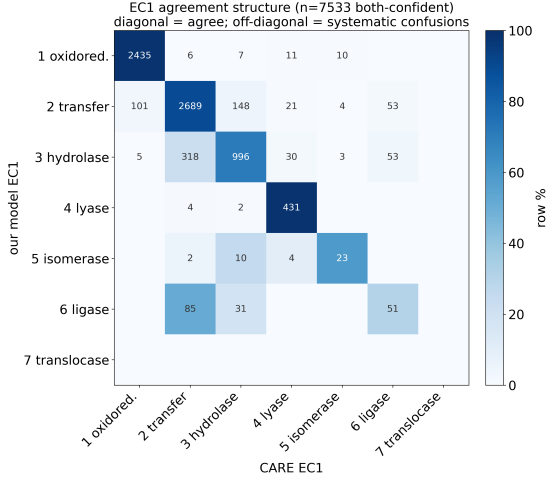
Each rung is one controlled change on the same harness — so "beats SOTA" is decomposable and defensible.



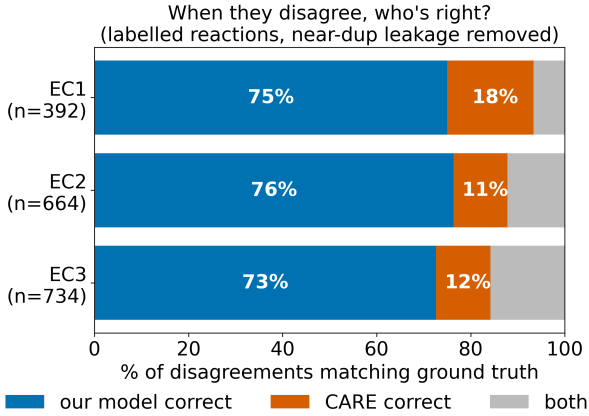


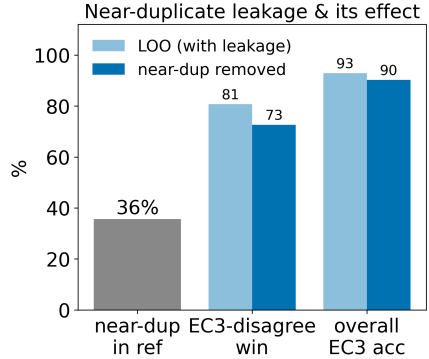
Our retriever (ModelSEED ref, ECFP4-diff‖DRFP) vs CARE's Similarity baseline (curated 62k ref, DRFP), same 15-NN protocol, on the 11.5k reactions we score confidently.

An *independently-built* predictor corroborates us, and **agreement calibrates**: EC3 62%→**95%** as both gate up. **5,721 reactions** get a *double-validated* EC subclass. Where the two disagree on



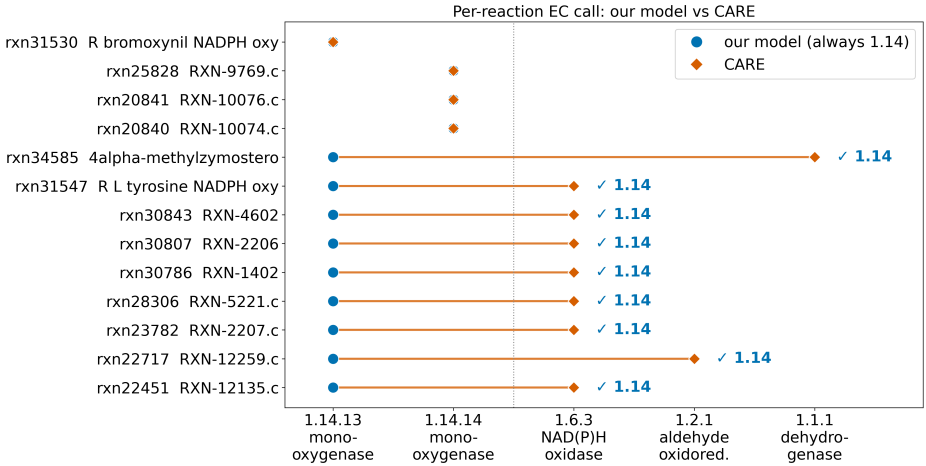
88% on the diagonal; the ∼12% off-diagonal is *systematic* — transferase↔hydrolase (318+148, a hard boundary), transferase↔oxidoreductase (101), ligase→transferase (85). And on labelled reactions (ground truth), **when they disagree our model is right ∼73–80%** vs CARE ∼10% — but that is a retriever on its *own* ModelSEED reference (near-dup leakage + home-DB), so it's the





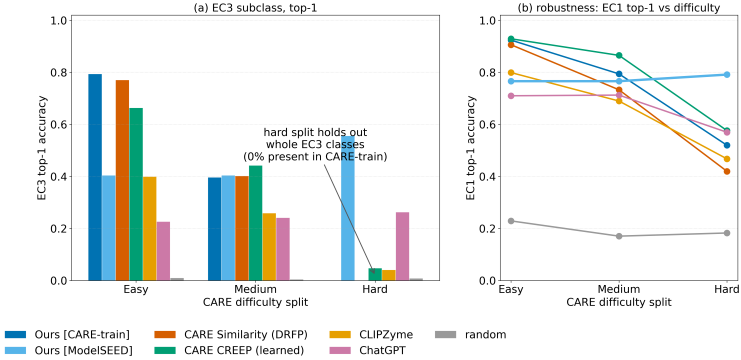
On the 13,721 *labelled* reactions (ModelSEED EC = truth); our model LOO 15-NN over the same set, CARE over its own 61,766 ref. Disagreements only.

On ModelSEED ground truth, when they disagree **our model matches it ∼73–76%** vs CARE ∼12–18%. But 36% of the set has a near-duplicate twin (LOO leakage); removing it deflates the win 81→73 and acc 93→90 — and it's still a retriever on its *own* ModelSEED reference. Trust it *for ModelSEED*, not as a general win.



The 13 O_2-monooxygenase corrections the balance tool made (*not* a disagreement sample — 4 of the 13 agree). Bottom 9: our model holds **1.14** (monooxygenase); CARE scatters to **1.6.3 oxidase / 1.1.1 dehydrogenase**.

On this *cofactor-dominated slice*, reading the balanced reaction (O_2+NAD(P)H→H_2O, substrate oxidized) confirms **1.14 on every dispute** — CARE is pulled off by the shared NAD(P)H fingerprint.



CARE task-2 reaction→EC splits — ground truth Rhea/SwissProt (CARE's home field, *ours away*). Our ECFP4-diff‖DRFP retriever with **CARE's own train** (dark) and **ModelSEED** (light) as reference, vs CARE's published baselines, CARE's own top-k metric. Leakage-checked: exact test∩ModelSEED = 0/0/1%.

Off the home field the 73-vs-11 dominance *evaporates* — honestly. Same reference, we **edge CARE's Similarity** (EC1 $+2/+6/+10$pp) but only *tie* the learned **CREEP**. Every train-restricted retriever falls to ∼0 EC3 on *hard* (whole classes held out) — the Part-V sparsity ceiling, confirmed externally. Real edge = **coverage**: our ModelSEED tool stays **flat (EC1 0.77→0.79)** across difficulty while others collapse, spanning 98% of classes (no leakage).

1. **Structural difference fingerprints** + **agglomerative-average** are the strong, cheap baseline (EC3 AMI ≈0.63); **learned reaction embeddings** *underperform* them.
2. **Retrieval beats clustering** for annotation: confidence-gated 15-NN gives EC3 at 0.90 accuracy over 76% of reactions.
3. **CGR** (atom-mapped bond changes) wins the full-EC retrieval it was designed for, but is coverage-limited and not dominant alone.
4. **Multi-view fusion** of complementary views (**PhysChem⊕CGR⊕ChemBERTa**) **beats every individual method** — EC3 AMI 0.71 — unsupervised.
5. **More exotic** $\neq$ **better**: 3D shape, pharmacophore, learned-graph, and a 335M biochemistry-native model add no EC signal the 2D views lack (Part III).
6. On **neutral external ground truth** (CARE task-2, Rhea/SwissProt) the method is **on par** with the best baselines — the home-DB "73 vs 11" does not generalise; its real edge is **reference coverage** → graceful degradation on novel chemistry.
7. Next: **hierarchical-EC metric learning**, evaluated zero-shot on held-out EC subclasses.